AEROLAB

Faculty Demonstration & Graph Walkthrough

The one-sentence takeaway

AeroLab connects ESP32 control, sensors, a dashboard, and software fault simulations. Today's graphs support model/simulation claims; independent hardware validation is the next research step.

Suggested 5-minute faculty walkthrough

Time	Show / explain	Suggested wording
0:00–0:40	Dashboard source: DEMO, simulation, or live.	I'll distinguish simulation from physical telemetry.
0:40–1:25	MATLAB → ESP32 → sensors/actuator → dashboard.	The contribution is an integrated, auditable teaching demonstrator.
1:25–2:10	US1 fixed, US2 servo-mounted; echo validity and raw LDR.	This is angular-sector proximity response, not continuous hand pose.
2:10–3:15	Step plot: linear models versus clipped response.	This is a model benchmark, not measured servo performance.
3:15–4:05	Observer/residual/CUSUM with injected-fault marker.	The offset is injected in software; residual is not independent sensor validation.
4:05–5:00	State the next validation experiment.	I'll measure servo angle independently and report repeatable timing/detector statistics.

Terminology guardrails

Say	Avoid until measured
Model-linked demonstrator	Validated digital twin / flight controller
Servo ultrasonic sector scan	Continuous 2-D hand tracking
Software-injected drift	Hardware fault-detection validation
Raw, uncalibrated LDR ADC	Lux / calibrated photometry
Nominal 20 Hz loop	Verified end-to-end timing
Simulated parameter sweep	Guaranteed hardware robustness

How to read each graph

Graph / scope	What to point out	What it means / caveat
Step-response comparison	Setpoint and continuous, discrete-linear, and saturated curves.	Compare rise, overshoot, settling, and input bounds. Saturated trace is closer to firmware command logic, but remains model-only; do not quote it as measured servo dynamics.
Twin/model-state coincidence	Overlapping x_1 and twin-state traces; divergence when software offset is introduced.	Internal states share model propagation/control. Agreement is not independent validation of a physical plant.
Residual and CUSUM	Residual magnitude against δ; positive/negative accumulators against h; injection time.	Illustrates detector logic under a chosen synthetic fault/noise. It does not establish hardware false-alarm probability or detection latency.
Control effort and servo	Normalized u in $[0,1]$ and servo-angle command.	Shows commanded effort/angle mapping. Distinguish command from independently measured shaft position.
Observer estimation error	Position error and scaled velocity error decay in simulated run.	Model observer behavior only; no physical reference measurement is present.
Bode / root locus	Frequency response and nominal model pole movement under plotted assumptions.	Useful design analysis for the assumed model; not an empirical stability margin or robustness certificate for the hardware.
Parameter sweep	Spread of model responses under the stated parameter perturbations.	Sensitivity study on simulated plants. Do not state as measured reliability or a guarantee.
Attitude / sensor scopes	Roll/pitch, vibration, distance, temperature/gas, raw LDR.	Live only when source and freshness indicate hardware. LDR is ADC counts until calibrated; ultrasonic channels require fresh valid echoes.

Dashboard cleanup before the meeting

- Keep the source indicator visible; never present DEMO/simulation as LIVE.
- Use corrected analytics: assumed model parameters, firmware-matched gains, and no unsupported FAR/robustness claims.
- Call MATLAB views model analysis. Label residual/CUSUM as software-model injected simulation.
- For hardware demo, verify fresh packet age and sensor validity; record CSV. Connection alone is not validation.
- Describe hand mode as sector scan/lock; verify US2 echo, servo motion, reacquisition, alarms, and limits.
- Call LDR/MQ raw or uncalibrated unless calibration and warm-up are documented.

Likely faculty questions

Question	Short answer
Is it a digital twin?	Not yet validated: residual uses internal model states plus injected drift, not an independent actuator measurement.
Does it track a hand?	It scans and locks/reacquires a range sector; one sensor cannot recover continuous 2-D hand pose.
Where did ω_n, ζ come from?	They are design assumptions; identify them from measured servo/load data.
What does the alarm prove?	Software response to coded synthetic offset only; hardware rates remain unknown.
Strongest result?	Integrated model-to-firmware-to-dashboard teaching demonstrator.
Next experiment?	Measure shaft angle independently, synchronize logs, identify/validate model, test held-out faults.

A strong next validation package

Validation area	Measure before a performance claim
Hardware provenance	Board and firmware version, supply voltage, wiring, sensor identity, environment, calibration procedure, and test date.
Independent actuator ground truth	Encoder or optical angle reference synchronized with command and telemetry timestamps.
Model identification	Repeated step/frequency tests; fit residuals and validation on separate runs, including deadband, rate limits, and saturation.
Runtime timing	Packet age, missed packets, loop-period/jitter distribution, command-to-measurement delay, and reset/error logs.
Fault detector	Baseline/fault scenarios; false alarms per hour, detection probability vs fault magnitude, delay distribution, misses, sample count, and confidence intervals.
Photonic sensor	LDR ADC-to-lux calibration against a reference meter at multiple levels; document geometry and ambient-light control.
Hand interaction	Define target distance/angle/speed and score sector detection/reacquisition. Continuous 2-D tracking needs adequate additional sensing geometry.